

Module 6 Introduction	
Lesson 1: Bias, Variance and Model Complexity	
Lesson 2: Bayesian Approach and BIC	
Reading: Bayesian Approach and BIC Reading	4h
Video: AIC and BIC	4 min
Video: Minimum Description Length (MDL)	7 min
Practice Assignment: Bayesian Approach and BIC Quiz	Unlimited
Lab: Coding Example	1h
Lab: Coding Exercise	1h
Lesson 3: Vapnik-Chervonenski (VC) Dimension	
Lesson 4: Cross Validation	
Lesson 5: Bootstrapping	
Module 6 Summative Assessment	
Module 6 Summary	

Your grade: 100%

You lasted: 100% • View history

Bayesian Approach and BIC Quiz

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Bayesian Approach and BIC Quiz

12 questions

1. In the context of AIC and BIC, which model has a lower value typically indicate?

- ☐ A simpler model that is likely to overfit the data.
- ☒ Higher prediction accuracy on new data.
- ☐ A more complex model with more parameters.
- ☐ A better balance of model fit and complexity of fits.

☒ Correct ☐ Submitted ☐ Attempts ☐ Let's chat

Correct Lower AIC or BIC indicates a better trade-off balance between complexity and fit.

[Try again](#)

2. What does the AIC specification penalize model selection?

- ☐ The error rate of the fitted model.
- ☒ The computational cost of the model.
- ☐ The amount of data used to train the model.
- ☐ The number of parameters in the model.
- ☐ The duration of time taken to fit the model.

☐ Line ☒ Scatter ☐ Report as issue

☒ Correct ☐ Correct ☐ Correct ☐ Correct

Correct AIC penalizes the number of parameters to avoid overfitting.

3. What is the central idea of the Minimum Description Length (MDL) principle in model selection?

- ☐ It selects the model with the lowest prediction error.
- ☐ It chooses the model with the minimum number of parameters.
- ☒ It seeks to find the model that provides the shortest combined representation of the model and the data.
- ☐ It emphasizes on minimizing the computational time required for modeling.

☒ Correct ☐ Correct ☐ Correct ☐ Correct

Correct MDL is based on the idea of selecting the model that minimizes the total length of describing the model and the data.